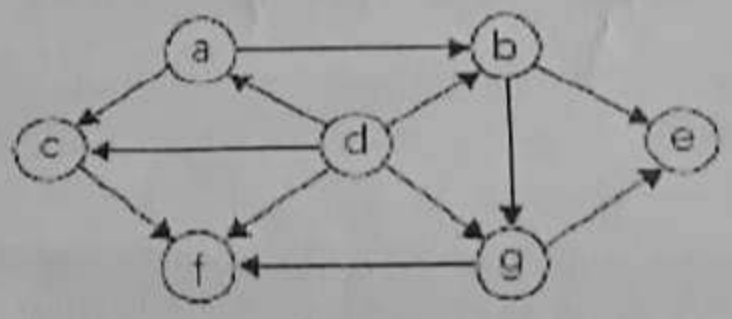
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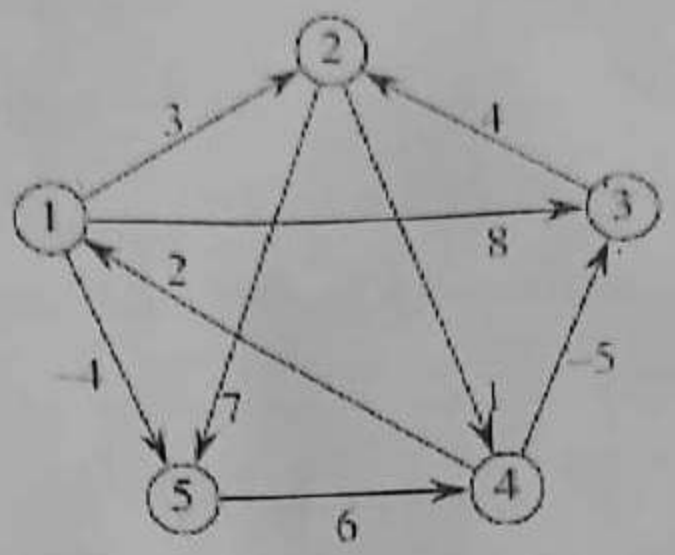
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End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Programme: <B.Tech (CSE-AI/ IT/AI&ML/ECE-AI)> Semester:<4th>
(May, 2026)

Subject Code: BCS 201	Subject: Design and Analysis of Algorithm
Time: 3 Hours	Maximum Marks :60

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1		(5*4=20)
	a) Discuss the Big-O and Big-Omega asymptotic notations and their significance in analyzing algorithm. Also prove that ½ n(n-1) = $\Theta(n^2)$ by finding the values of C_1, C_2, and n_0.	5
	b) Why is Huffman coding optimal? Construct the Huffman Tree and calculate the total number of bits required to encode the message for the following set of characters with their frequencies: A:5, B:9, C:12, D:13, E:16, F:45	5
	c) Find the topological sort of the following graph.  	5
	d) Construct the string-matching automaton for the pattern P= **aabab** and illustrate its operation on the text string T= **aababababaabab**.	5
	UNIT I	
Q2	Solve the following Recurrences i. $T(n) = 4T(n/2) + n^2\sqrt{n}$ ii. $T(n) = \sqrt{n}T(\sqrt{n}) + n$	(5,5)
Q3	Write an algorithm to find both the maximum and minimum elements in an array simultaneously using a divide-and-conquer approach, with a suitable example. Also, write its recurrence relation and analyze the number of comparisons required.	(10)
	UNIT II	
Q4	Differentiate between the dynamic programming and greedy approach. Write an algorithm and analyze its time complexity to solve the Matrix Chain Multiplication problem, and determine the optimal parenthesization for the following matrix chain: **A1(5×4), A2(4×6), A3(6×2), A4(2×7)**	(10)
Q5	How is the principle of optimality used in dynamic programming? Write an algorithm to find the Longest Common Subsequence and analyze its time and space complexity by constructing the table for the strings X="ABCBDAB" and Y="BDCABA".	(10)

	UNIT III	
Q6	For the given weighted directed graph, apply the Floyd–Warshall algorithm to find the shortest paths between all pairs of vertices. Construct the initial distance matrix and show the intermediate updates leading to the final shortest path matrix.  Also, write the algorithm and analyze its time complexity.	(10)
Q7	Explain the N-Queen problem and write an algorithm to solve it using the backtracking approach. Analyze the time complexity of the algorithm and for N=4, show all the steps involved in the backtracking process, including the state-space tree or partial configurations.	(10)
	UNIT IV	
Q8	Explain the Rabin–Karp string matching algorithm and analyze its time complexity. Working modulo q =11, how many spurious hits does the Rabin-Karp matcher encounter in the text T =14159265 when looking for the pattern P= 26?	(10)
Q9	Discuss the classes P, NP, NP-hard, and NP-complete with a suitable diagram, and describe the steps involved in proving that a problem is NP-complete.	(10)